Agenda

* Overview of Cars24 Data
* Goal of ML : Generalization
* Linear Regression Intuition
* Implementation using Sklearn
* Evaluation Metric

✓ Assignments

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Recap

→ classification, regression, clustering, recommendation, forecasting

✓ Supervised learning => X, y < classification, Regressions

✓ unsupervised learning =>



What do you think about the nature of Car Resale price prediction?

5 users have participated

✓ A	Regression	100%
B	Classification	0%
C	Clustering	0%

End Quiz Now

Made with Goodnotes

Cars 24 Problem Overview

Data scientist at **Cars 24** → Sells pre-owned cars

Resale price

→ To automate pricing the old car



Given car features □□□□□□□□□□

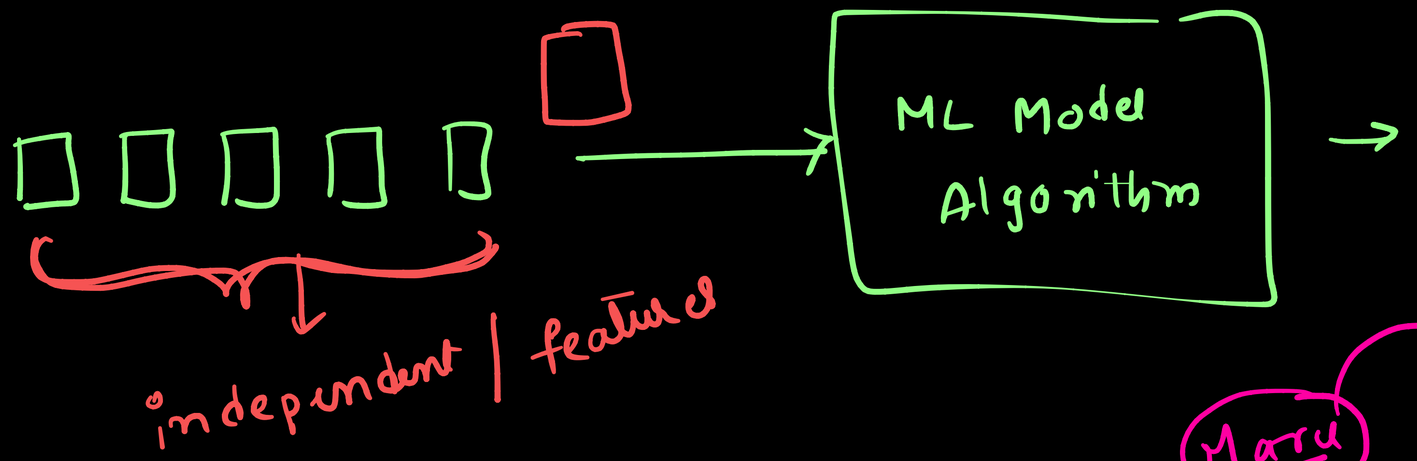
(Make, Model Mileage , Odometer, Service History etc.) →

selling price

↓

Predictors / Features

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Maruti
Hyundai
Ford

OHE

Target Encoding

Make	Selling Price
Maruti	1.2
Hyundai	5.5
Hyundai	2.15
Hyundai	2.26
Ford	5.70

Target Encoding

MEAN
Maruti - 1.2
Hyundai - 3.3
Ford - 5.70

Replace

Make	Selling Price
1.2	1.2
3.3	5.5
3.3	2.15
3.3	2.26
5.7	5.70

Target encoding replaces the categories with a number representing the average target value associated with each category.

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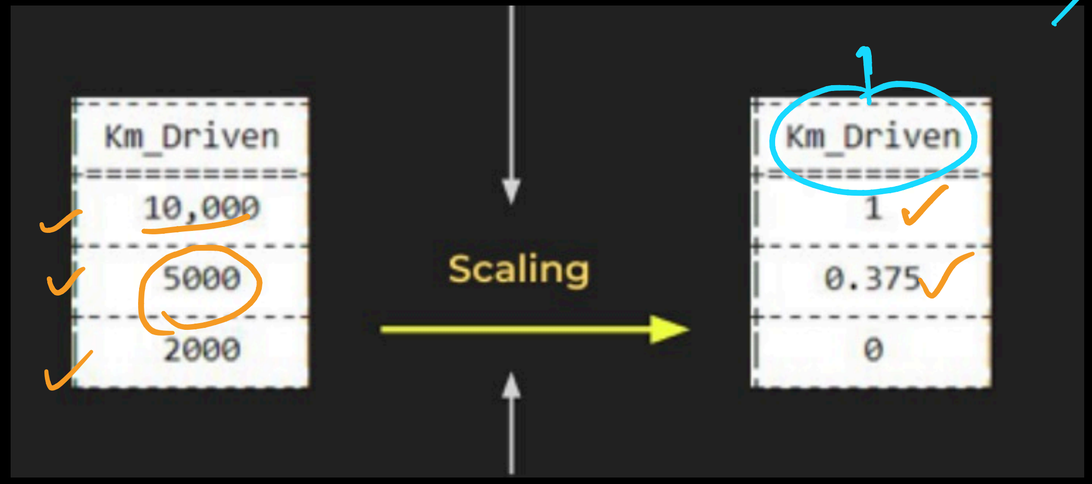
MinMax Scaling / Normalization

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

$$\frac{10000 - 2000}{10000 - 2000} \Rightarrow 1$$

$$\frac{5000 - 2000}{10000 - 2000} = \frac{3000}{8000} = 0.375$$

0 to 1

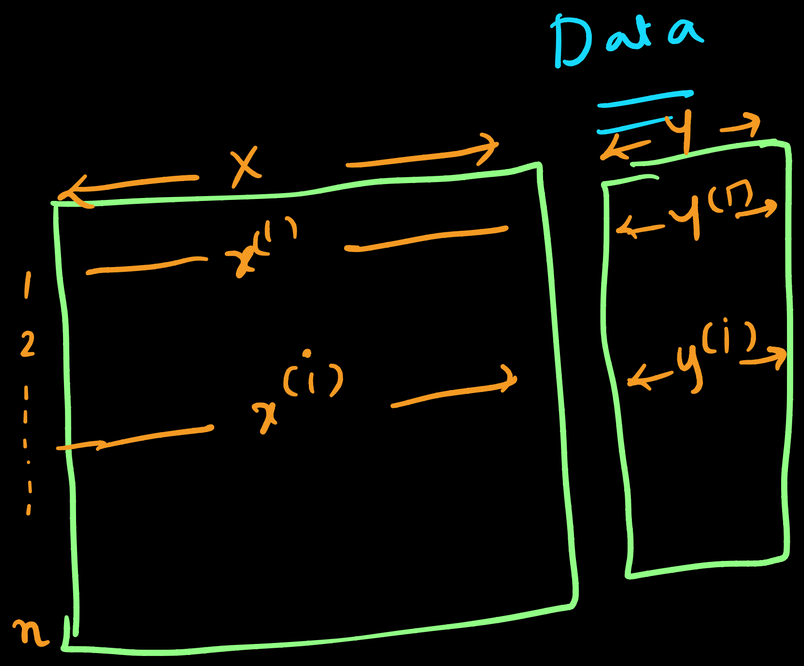


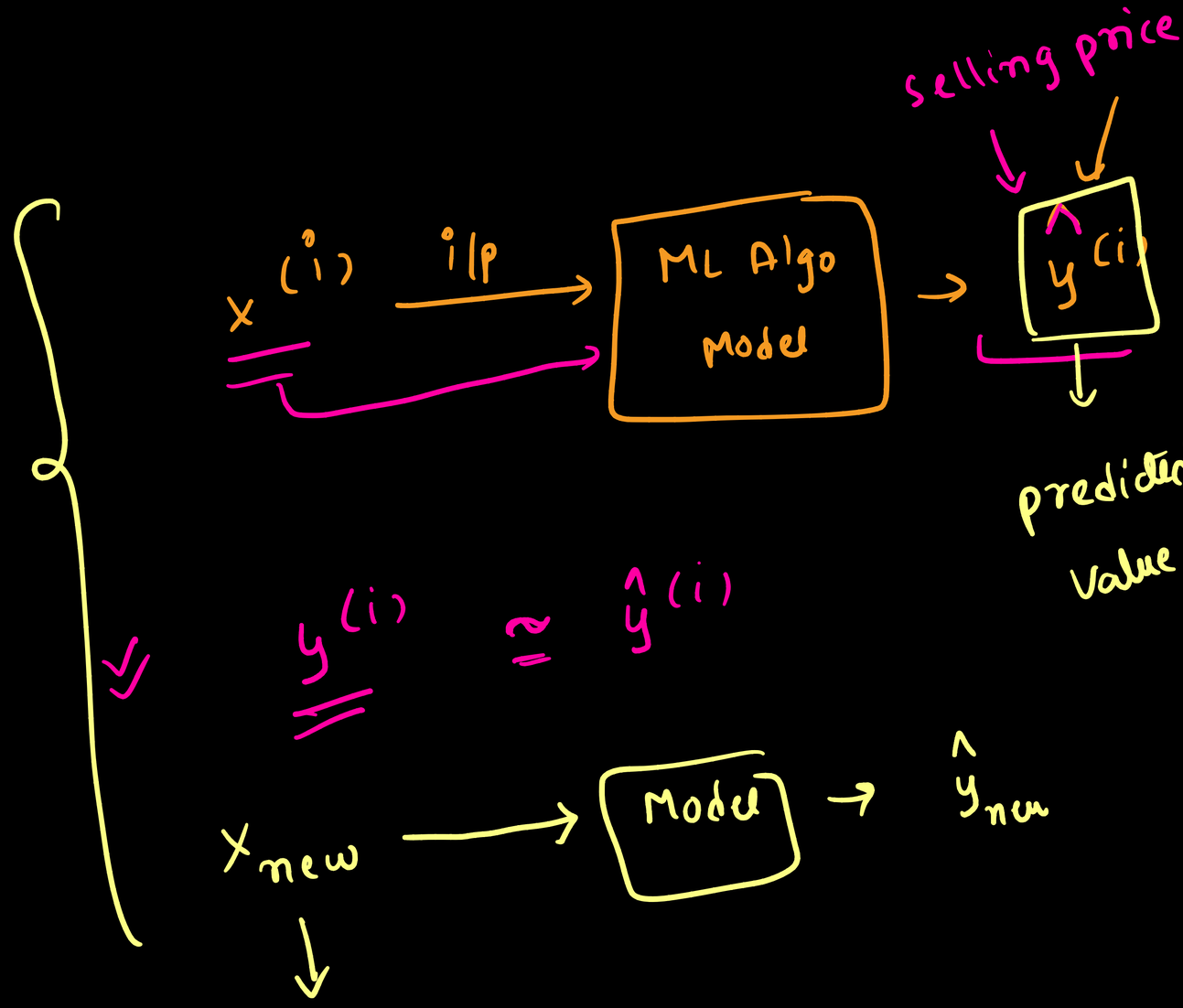
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How do you think we should handle the large number of categories in make and model column?

30 users have participated

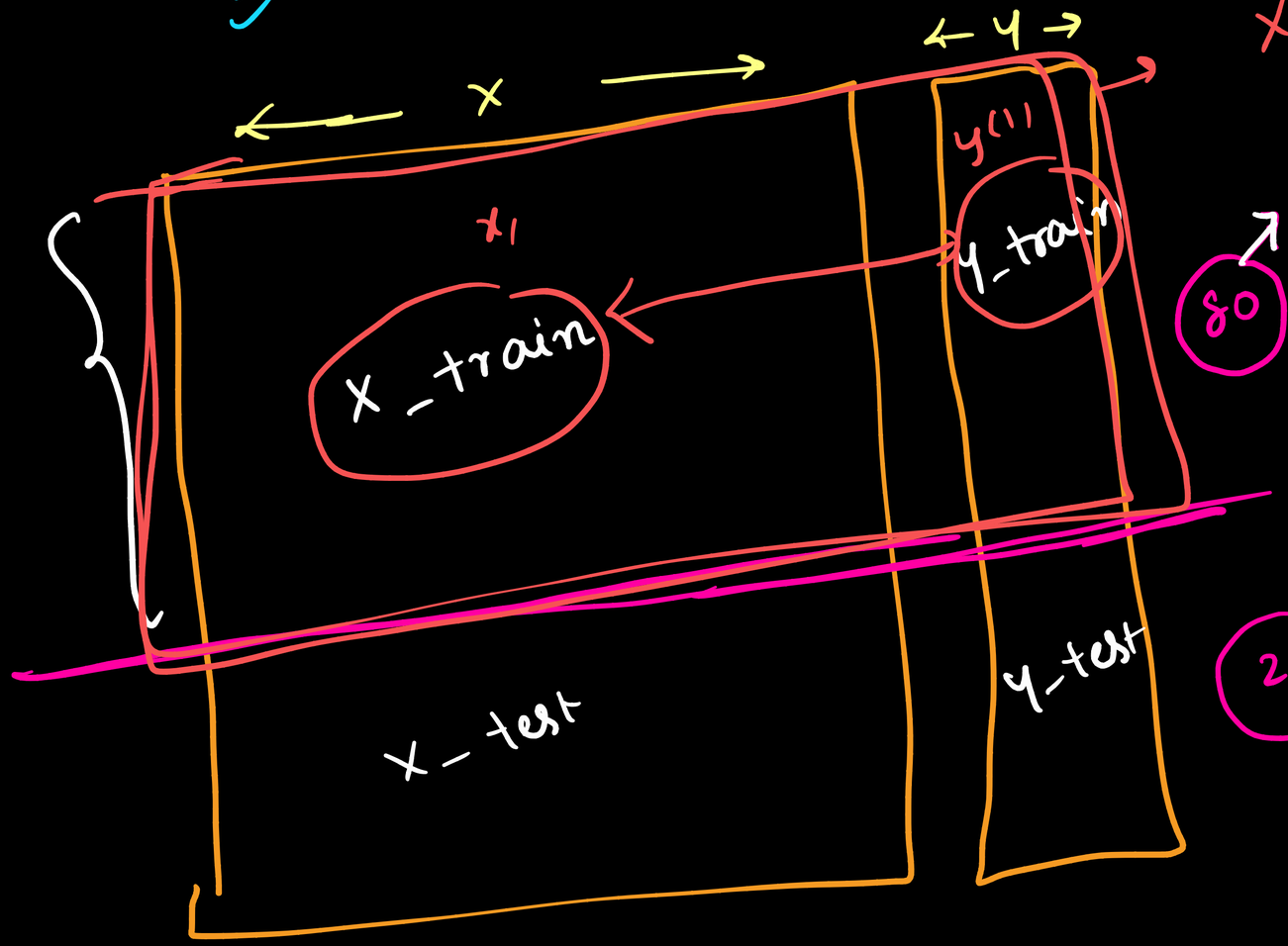
A	One Hot Encoding	20%
B	Label Encoding	3%
✓ C	Target Variable Encoding	77%





1. Training phase

2. Testing phase



X_train, y_train

Train Dataset

train my model

Test Dataset

model is doing good(or) not with an unseen data

Intution of Lin. Reg

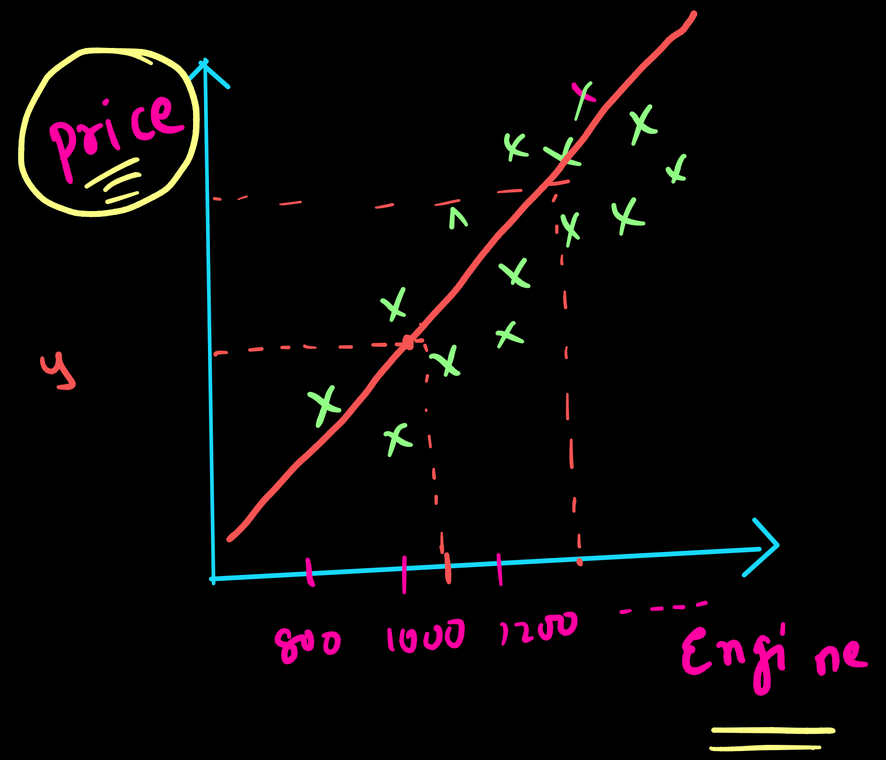
- Univariate Lin Reg [1 - feature]
- Multivariate Lin Reg [>1 - features]

Univariate Lin Reg

Engine → price of the data

engine = 1200

$x^{(i)}$ → line f() → $\hat{y}^{(i)}$



Eqn of straight line

$$y = mx + c$$

$$g = w_1 x + w_0$$

- x → independent variable → engine
- w_1 → weight / coefficient
- w_0 → intercept

New car → 1000 cc → what is the price

Assume

$$\hat{y} = 200\,(x) + 50000$$

(w_1 = 200, w_0 = 50000)

$$\Rightarrow 200 * 1000 + 50000$$

2.5 Lacs

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Evaluation Metrics

$x^{(1)} \rightarrow y^{(1)} - \hat{y}^{(1)} \rightarrow e^{(1)}$

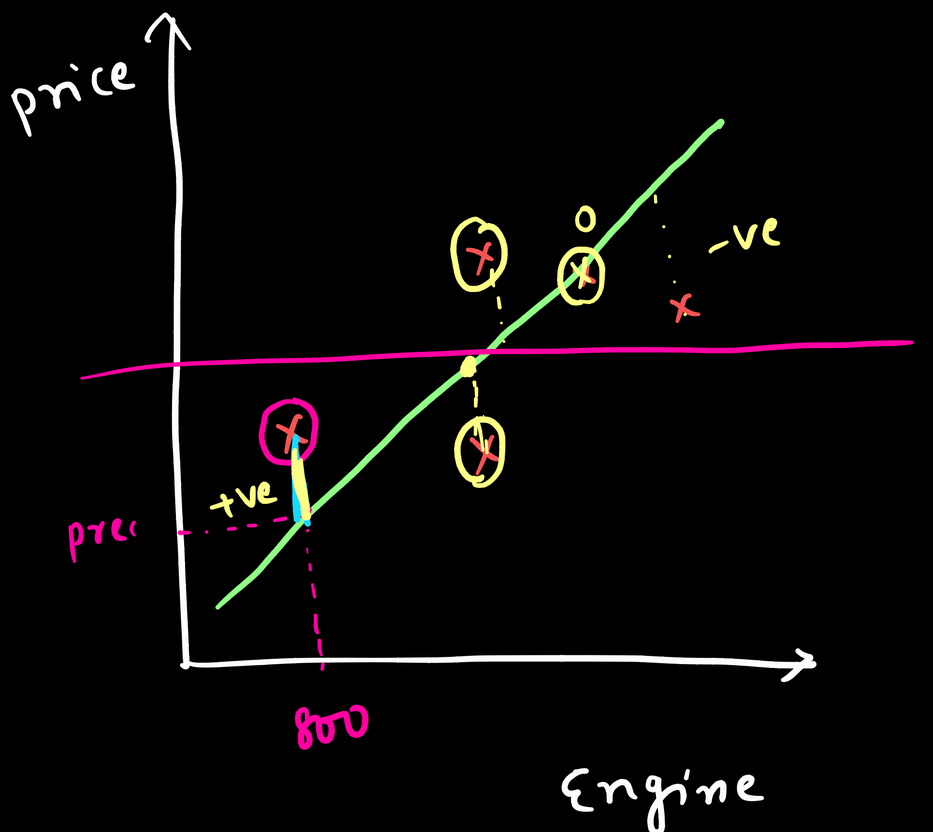
$x^{(2)} \rightarrow y^{(2)} - \hat{y}^{(2)} \rightarrow e^{(2)}$

$x^{(3)} \rightarrow y^{(3)} - \hat{y}^{(3)} \rightarrow e^{(3)}$

$x^{(4)} \rightarrow y^{(4)} - \hat{y}^{(4)} \rightarrow e^{(4)}$

$x^{(5)} \rightarrow y^{(5)} - \hat{y}^{(5)} \rightarrow e^{(5)}$

$$\text{Total error} = \sum_{i=1}^{n} e^{(i)}$$



$y^{(i)} - \hat{y}^{(i)}$

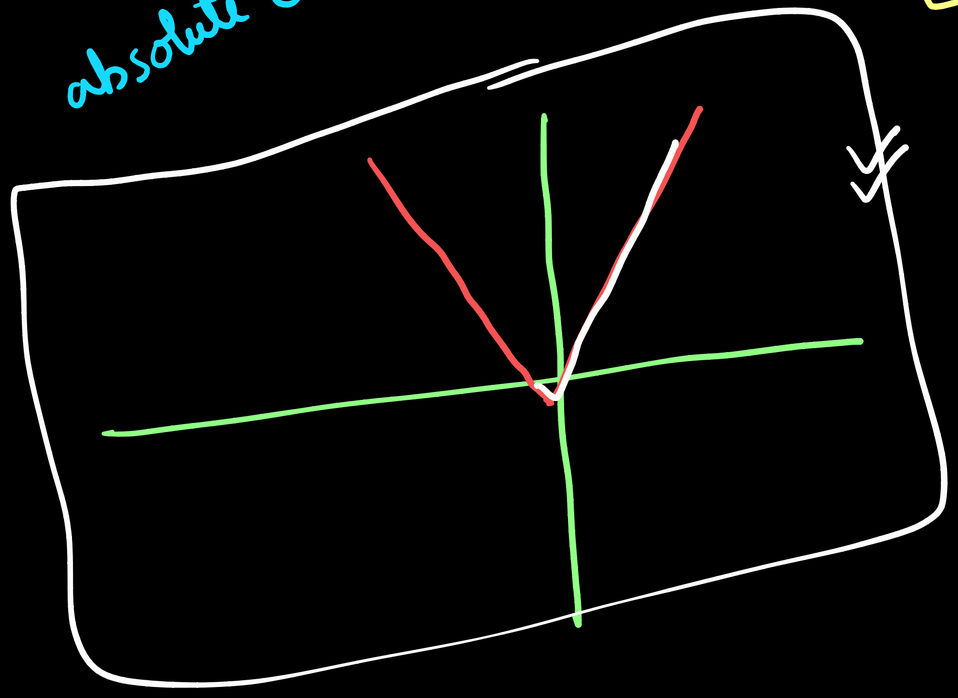
$$e_1 + e_2 + e_3 + e_4 + e_5 = 3$$

$5 \quad -4 \quad 3 \quad 0 \quad -1$

$\hookrightarrow \quad 13$

(1) (M) A E $= \frac{1}{n} \sum_{i=1}^{n} \left| y^{(i)} - \hat{y}^{(i)} \right|$

Mean absolute Error



$\Rightarrow$ it is not differentiable

$x = 0$

$x \to 0^+ \quad 0.99\ldots$

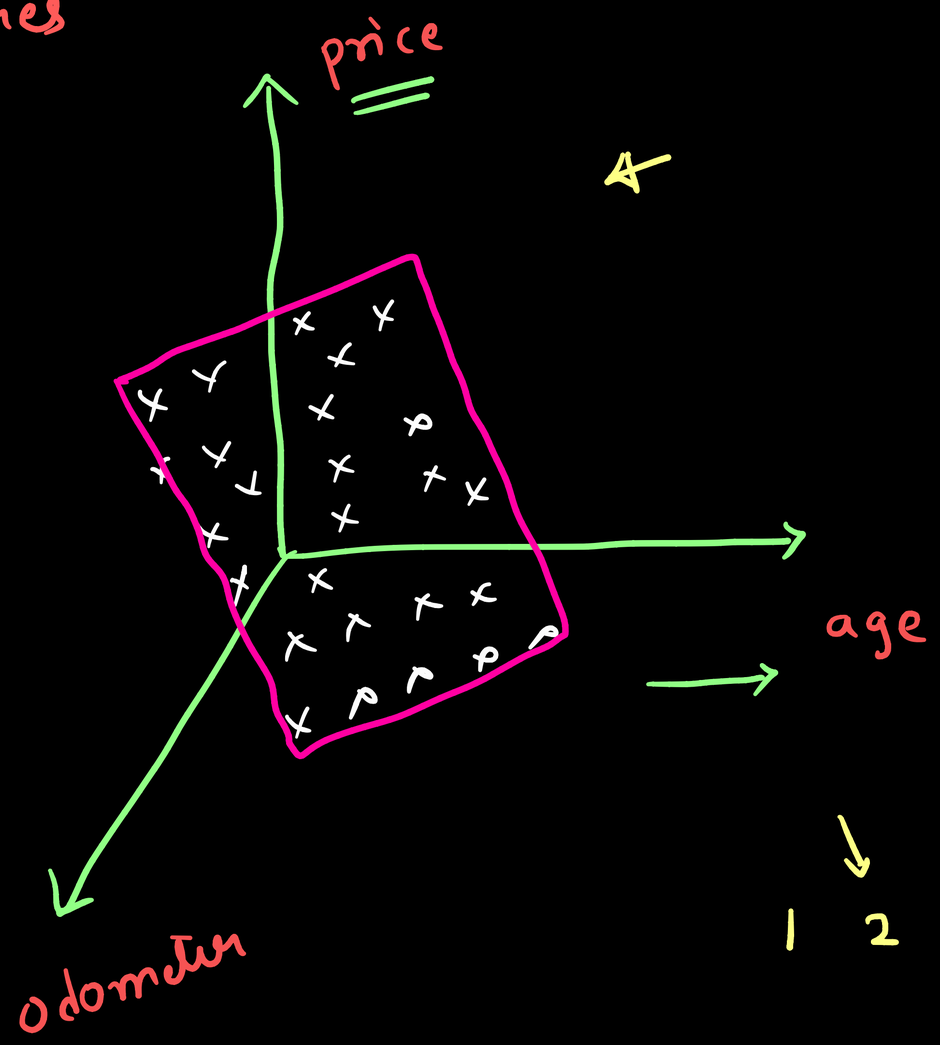
$x \to 0^-$

2- input features

Age → x_1

odometer → x_2

Age	odo meter	price



$$\hat{y} = w_1 x_1 + w_2 x_2 + w_0$$

weights, age, odometer

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0$$

$$= -10000\, x_1 - 10\, x_2 + 5000$$

d-features

$$w_1 x_1 + w_2 x_2 + \text{----} + w_d x_d + w_0$$

$$\vec{w}^T \cdot \vec{x} + w_0$$

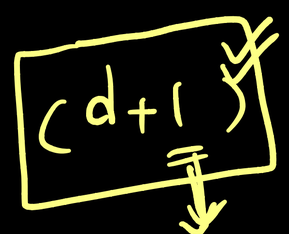
$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_d \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

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1 feature → 2D line

2 features → 3D

⋮

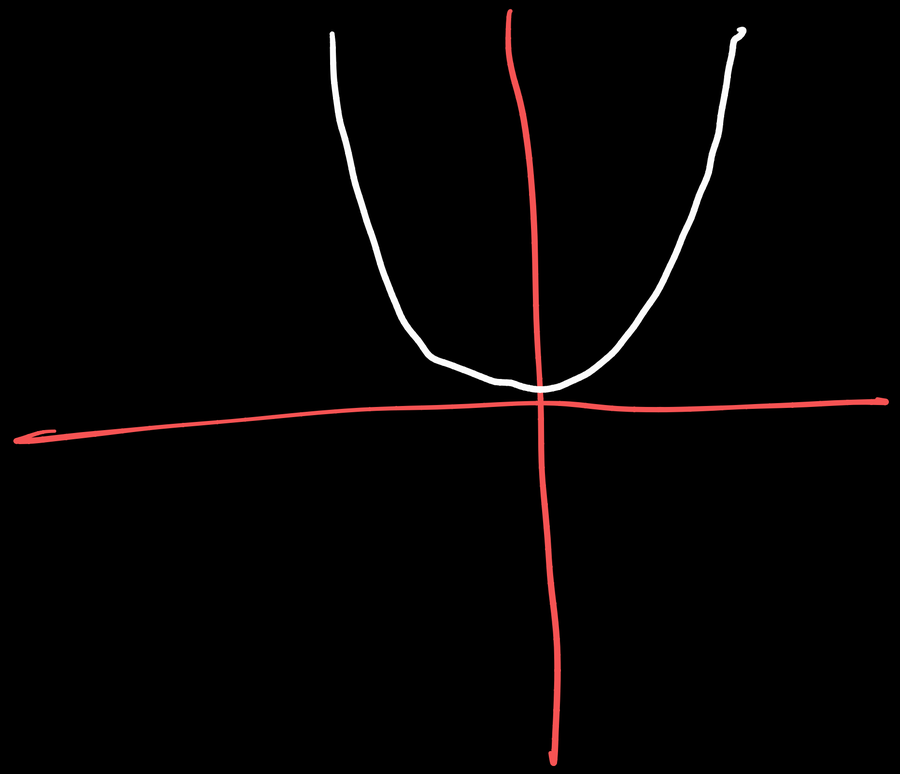
d features → 

label (price)

MSE

Mean Squared Error

$$= \frac{1}{n} \sum_{i=1}^{n} \left(y^{(i)} - \hat{y}^{(i)} \right)^2$$



	MSE
M_1	9.62
M_2	15.41

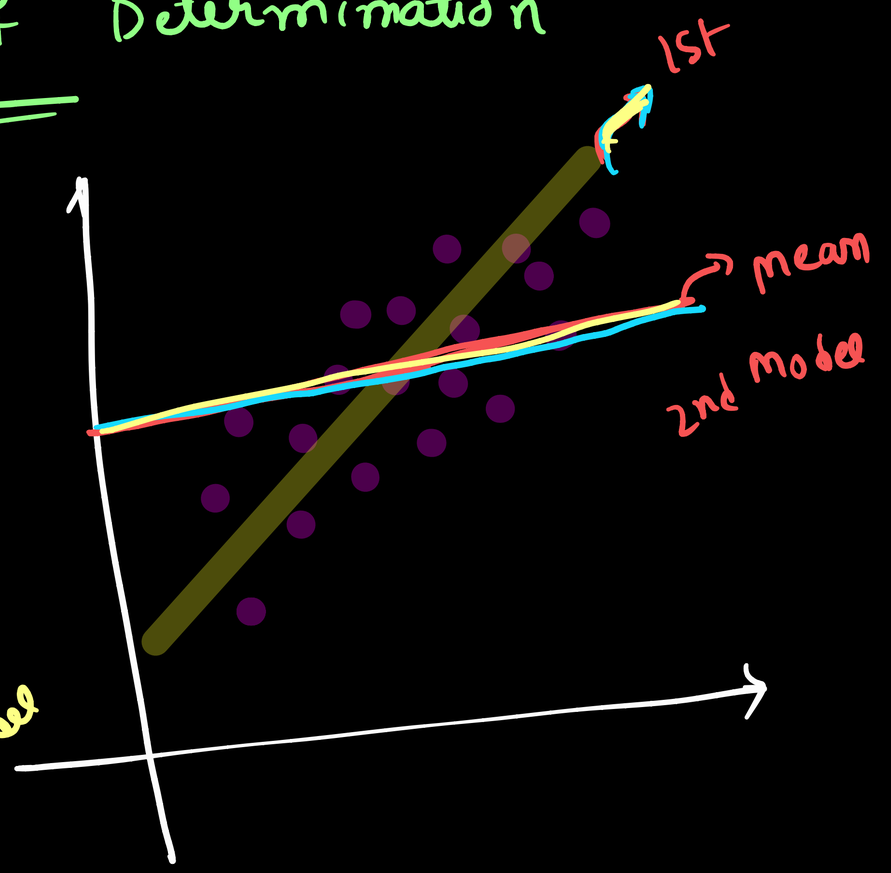
How to evaluate / model

R2-score / R-square / coeff of Determination

$$R2\ score = 1 - \frac{SS_{res}}{SS_{total}}$$

SS_{res} → MSE of good model

SS_{total} → MSE of mean model

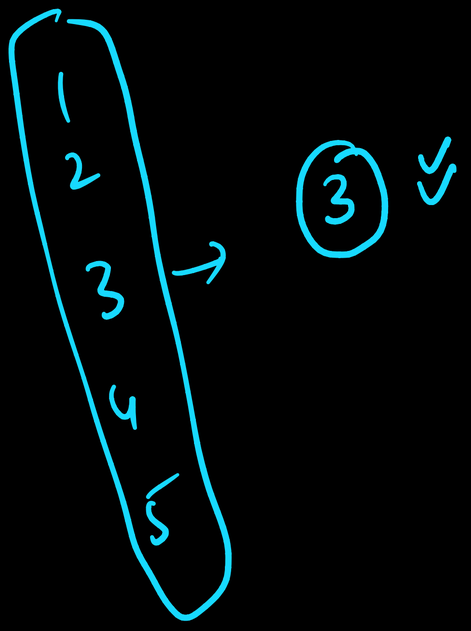


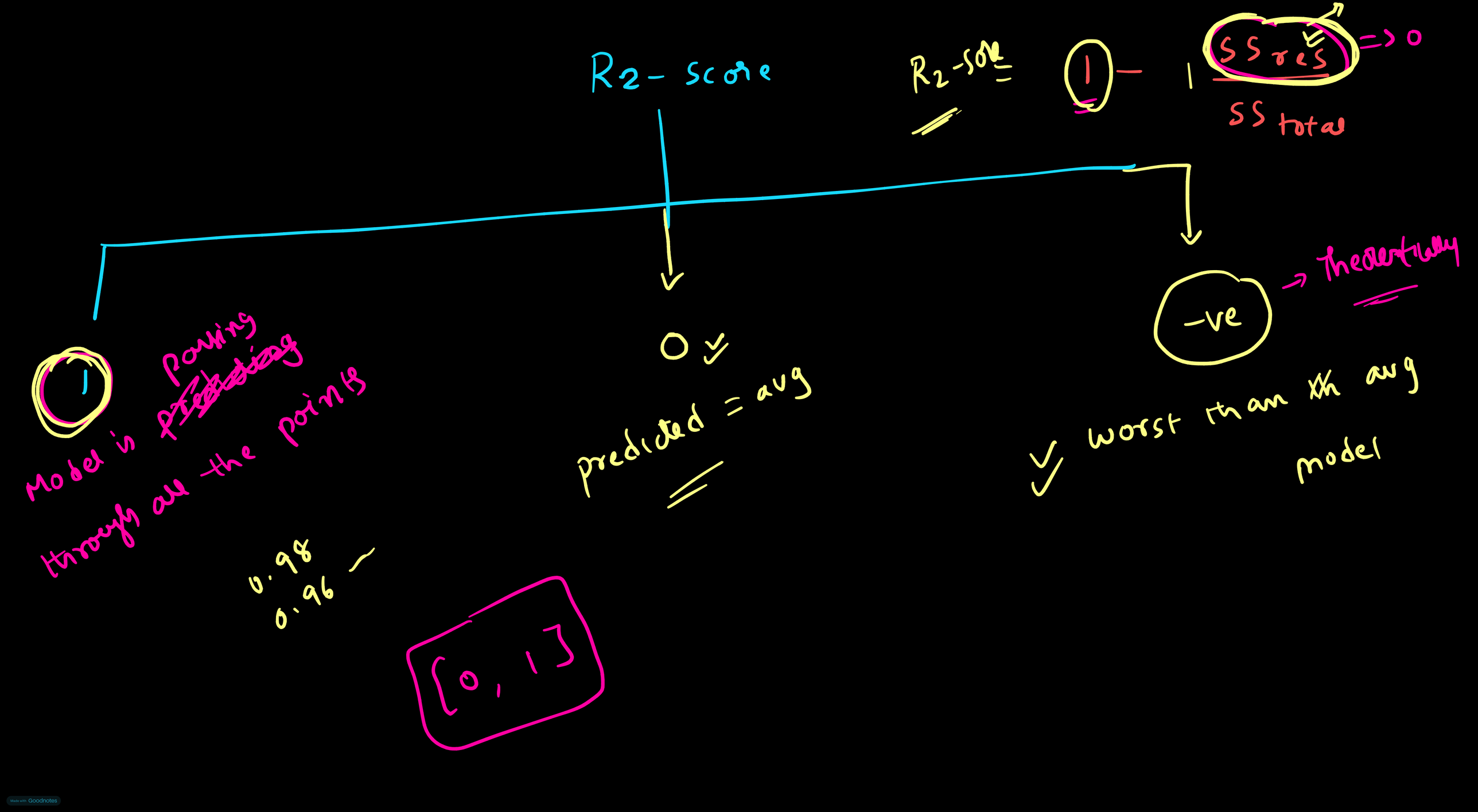
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$$= 1 - \frac{\frac{1}{n}\sum_{i=1}^{n}\left[y^{(i)} - \hat{y}^{(i)}\right]^2}{\frac{1}{n}\sum_{i=1}^{n}\left(y^{(i)} - \bar{y}\right)^2}$$

MSE




R2 - score
R2-score = 1 - SS res / SS total => 0
1
Model is Poorly fitting through all the points
0
predicted = avg
-ve
theoretically
worst than the avg model
0.98
0.96
[0, 1]

In linear regression, if the MSE value is 0, it indicates:

6 users have participated

A	The predicted values perfectly match the actual values.	100%
B	The model has no predictive power and fails to explain the dependent variable.	0%
C	The model has high bias and underfits the data.	0%
D	The model has high variance and overfits the data.	0%

End Quiz Now

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